NGINX WEBSERVER

NGINX is a high-performance, open-source web server software and reverse proxy, renowned for its speed, stability, and low resource usage. It handles high-traffic websites by managing numerous concurrent connections using an efficient event-driven architecture, often serving as a load balancer, reverse proxy, HTTP cache, and media streamer.

Key Features and Functions:

- **Web Server:** Efficiently serves static content (like CSS, images, and HTML) directly, while passing dynamic requests to application servers.
- **Reverse Proxy:** Sits in front of backend servers (e.g., Apache, Node.js, Python), receiving client requests and directing them to the appropriate server.
- **LoadBalancer:** Distributes incoming traffic across multiple backend servers to ensure high availability and prevent single-point failures.
- **Caching:** Acts as a content cache to store frequently requested content, reducing load times and server strain.
- **Security:** Provides protection against DDoS attacks and acts as a web application firewall.

Why NGINX is Used:

- **Performance:** Designed to solve the C10K problem (handling 10,000+ concurrent connections).
- **Efficiency:** Uses a non-threaded, asynchronous, event-driven architecture that minimizes CPU and memory usage.
- **Flexibility:** Supports HTTP, HTTPS, TCP, UDP, and mail proxying (IMAP/POP3/SMTP).

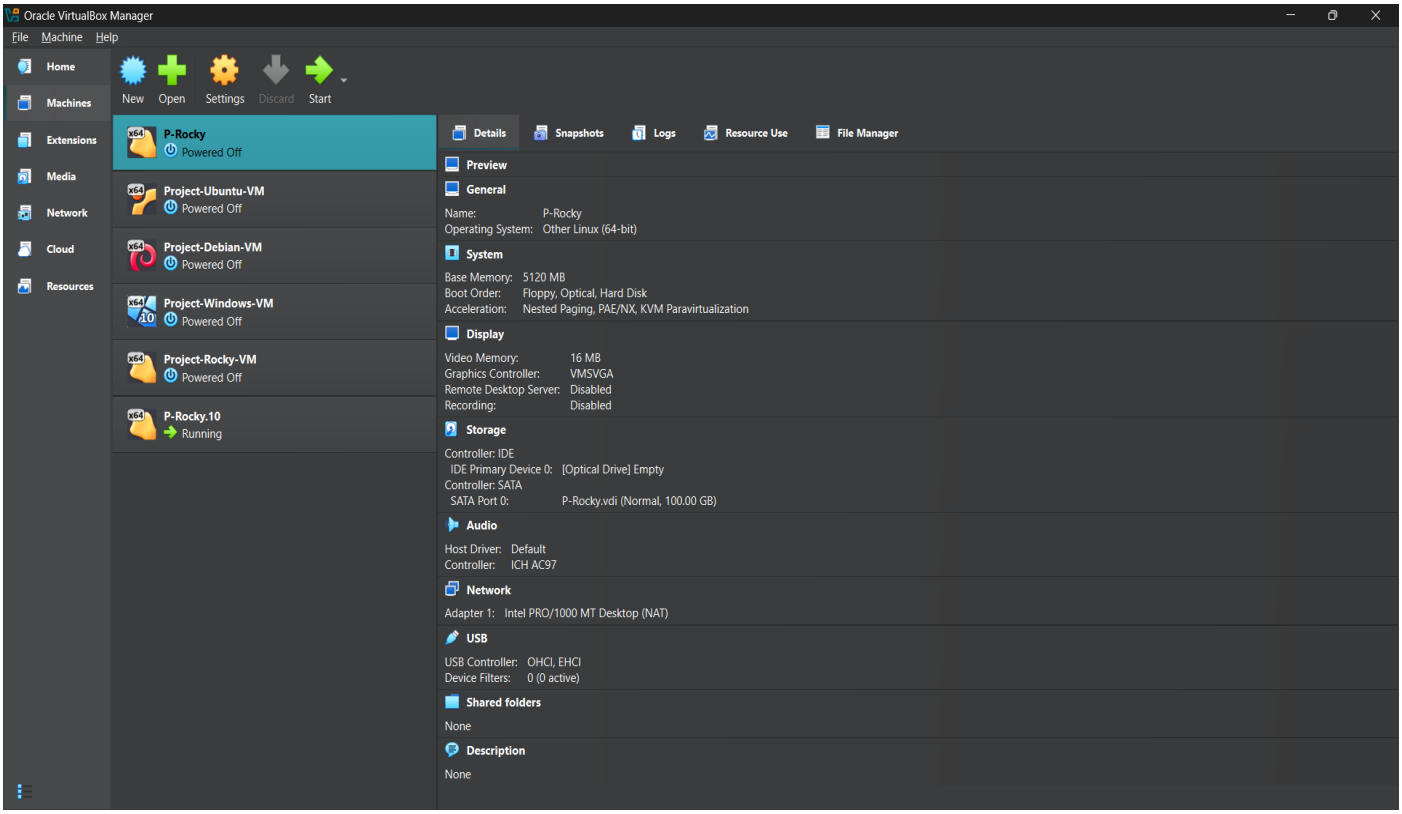
It is frequently used by major, high-traffic platforms like Netflix, Dropbox, and is a key component in containerized environments like Kubernetes.

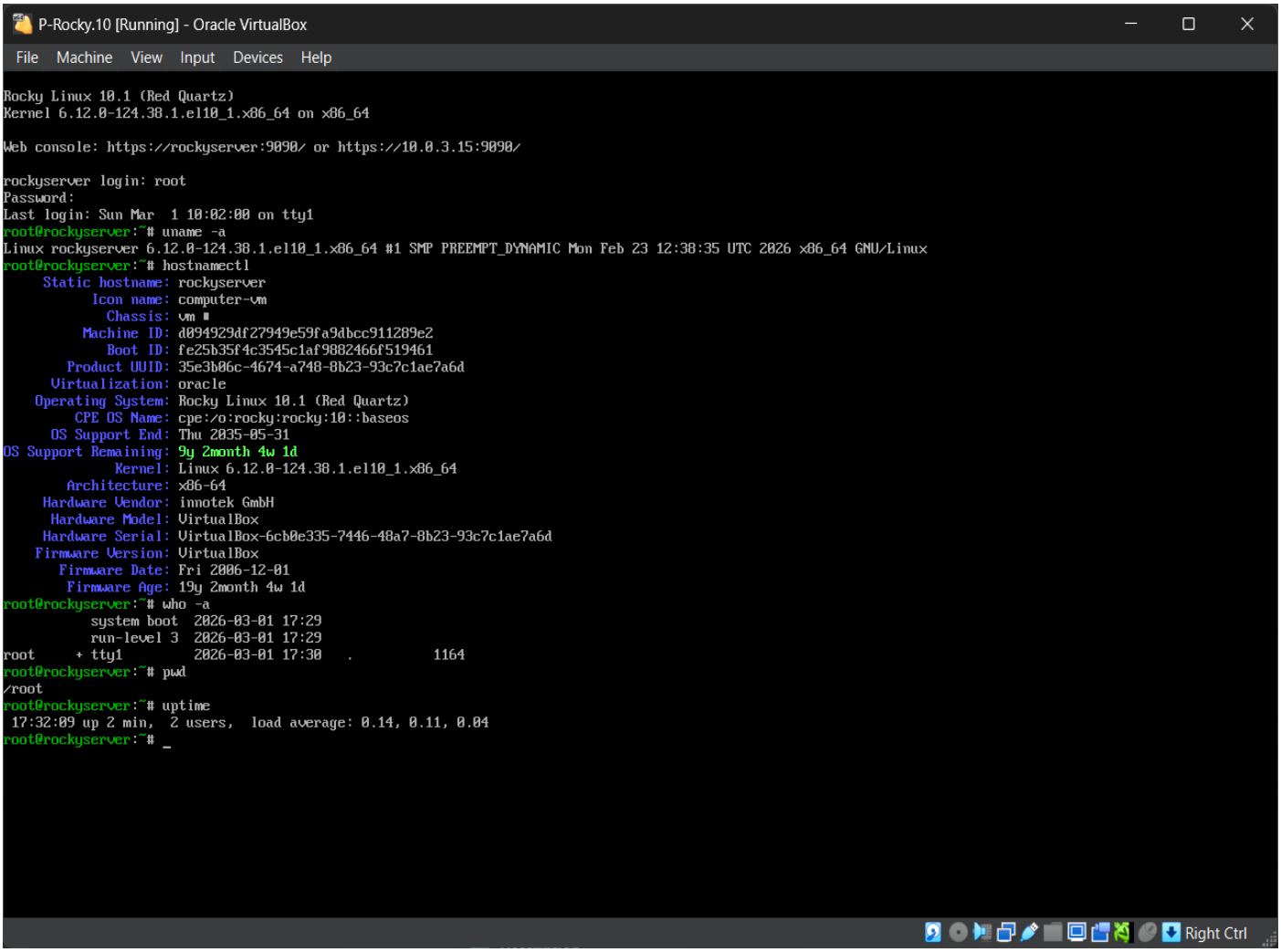
What is in this Document:

Title: Production Linux Server Setup & Automation

- ✓ Configuring and managing a production-style Linux server including
- ✓ User management
- ✓ Nginx web hosting
- ✓ Firewall setup
- ✓ Log monitoring via shell scripts
- ✓ Automating backups using cron jobs

Run a virtual machine in virtualbox and check its version and other software details using #hostnamectl command and #uname -a etc.,
then update all software packages using - #dnf update





Create a group named Developers add two users into it
use commands like #groupadd groupname , #useradd -aG groupname username,
check if users in group or not using #id username

P-Rocky.10 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
root@rockyserver:~# dnf update
Last metadata expiration check: 0:00:44 ago on Sun 01 Mar 2026 07:04:29 PM IST.
Dependencies resolved.
Nothing to do.
Complete!
root@rockyserver:~# groupadd Developers
root@rockyserver:~# useradd dinesh
root@rockyserver:~# passwd dinesh
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
root@rockyserver:~# useradd lokesh
root@rockyserver:~# passwd lokesh
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
root@rockyserver:~# usermod -aG Developers dinesh
root@rockyserver:~# usermod -aG Developers lokesh
root@rockyserver:~# id dinesh
uid=1001(dinesh) gid=1002(dinesh) groups=1002(dinesh),1001(Developers)
root@rockyserver:~# id lokesh
uid=1002(lokesh) gid=1003(lokesh) groups=1003(lokesh),1001(Developers)
root@rockyserver:~#
```

Add user in etc/sudoers , give root privileges
use - #nano /etc/sudoers, write username ALL=(ALL) ALL under root ALL=(ALL) ALL
then press ctrl+o → enter → crtl+x , this saves the edit.
Or add the user into wheel group , use - #usermod -aG wheel username

P-Rocky.10 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
# %:group are resolved via the group plugin by default.
# We enable always_query_group_plugin to restore old behavior.
# Disable this option for new behavior.
Defaults    always_query_group_plugin

Defaults    env_reset
Defaults    env_keep =  "COLORS DISPLAY HOSTNAME HISTSIZE KDEDIR LS_COLORS"
Defaults    env_keep += "MAIL PS1 PS2 QTDIR USERNAME LANG LC_ADDRESS LC_CTYPE"
Defaults    env_keep += "LC_COLLATE LC_IDENTIFICATION LC_MEASUREMENT LC_MESSAGES"
Defaults    env_keep += "LC_MONETARY LC_NAME LC_NUMERIC LC_PAPER LC_TELEPHONE"
Defaults    env_keep += "LC_TIME LC_ALL LANGUAGE LINGUAS _XKB_CHARSET XAUTHORITY"

#
# Adding HOME to env_keep may enable a user to run unrestricted
# commands via sudo.
#
# Defaults   env_keep += "HOME"

Defaults    secure_path = /sbin:/bin:/usr/sbin:/usr/bin

## Next comes the main part: which users can run what software on
## which machines (the sudoers file can be shared between multiple
## systems).
## Syntax:
##
##      user    MACHINE=COMMANDS
##
## The COMMANDS section may have other options added to it.
##
## Allow root to run any commands anywhere
root    ALL=(ALL)       ALL
lokesh  ALL=(ALL)       ALL

## Allows members of the 'sys' group to run networking, software,
## service management apps and more.
# %sys ALL = NETWORKING, SOFTWARE, SERVICES, STORAGE, DELEGATING, PROCESSES, LOCATE, DRIVERS

## Allows people in group wheel to run all commands
%wheel  ALL=(ALL)       ALL

## Same thing without a password
# %wheel        ALL=(ALL)       NOPASSWD: ALL

## Allows members of the users group to mount and unmount the
## cdrom as root
# %users  ALL=/sbin/mount /mnt/cdrom, /sbin/umount /mnt/cdrom

## Allows members of the users group to shutdown this system
# %users  localhost=/sbin/shutdown -h now
-- INSERT --                                                    101,28        95%
```

Right Ctrl

Now install nginx server - #dnf install nginx -y
start and enable nginx server - #systemctl start nginx, #systemctl enable nginx
check status using - #systemctl status nginx

```
root@rockyserver:~# dnf install nginx -y
Last metadata expiration check: 0:12:01 ago on Sun 01 Mar 2026 12:53:16 AM IST.
Dependencies resolved.
================================================================================================================================================
 Package                                  Architecture                    Version                                    Repository                          Size
================================================================================================================================================
Installing:
 nginx                                    x86_64                          2:1.26.3-1.el10                            appstream                           33 k
Installing dependencies:
 nginx-core                               x86_64                          2:1.26.3-1.el10                            appstream                          663 k
 nginx-filesystem                         noarch                          2:1.26.3-1.el10                            appstream                           11 k
 rocky-logos-httpd                        noarch                          100.4-7.el10                               appstream                           24 k

Transaction Summary
================================================================================================================================================
Install  4 Packages

Total download size: 730 k
Installed size: 1.9 M
Downloading Packages:
(1/4): nginx-1.26.3-1.el10.x86_64.rpm                                                                                176 kB/s |  33 kB     00:00
(2/4): rocky-logos-httpd-100.4-7.el10.noarch.rpm                                                                     338 kB/s |  24 kB     00:00
(3/4): nginx-filesystem-1.26.3-1.el10.noarch.rpm                                                                      43 kB/s |  11 kB     00:00
(4/4): nginx-core-1.26.3-1.el10.x86_64.rpm                                                                           989 kB/s | 663 kB     00:00
------------------------------------------------------------------------------------------------------------------------------------------------
Total                                                                                                                545 kB/s | 730 kB     00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                                          1/1
  Running scriptlet: nginx-filesystem-2:1.26.3-1.el10.noarch                                                                                  1/4
  Installing       : nginx-filesystem-2:1.26.3-1.el10.noarch                                                                                  1/4
  Installing       : nginx-core-2:1.26.3-1.el10.x86_64                                                                                        2/4
  Installing       : rocky-logos-httpd-100.4-7.el10.noarch                                                                                    3/4
  Installing       : nginx-2:1.26.3-1.el10.x86_64                                                                                             4/4
  Running scriptlet: nginx-2:1.26.3-1.el10.x86_64                                                                                             4/4

Installed:
  nginx-2:1.26.3-1.el10.x86_64     nginx-core-2:1.26.3-1.el10.x86_64     nginx-filesystem-2:1.26.3-1.el10.noarch     rocky-logos-httpd-100.4-7.el10.noarch

Complete!
root@rockyserver:~# systemctl start nginx
root@rockyserver:~# systemctl enable nginx
Created symlink '/etc/systemd/system/multi-user.target.wants/nginx.service' → '/usr/lib/systemd/system/nginx.service'.
root@rockyserver:~#
```

```
Install  4 Packages

Total download size: 730 k
Installed size: 1.9 M
Downloading Packages:
(1/4): nginx-1.26.3-1.el10.x86_64.rpm                                                                                176 kB/s |  33 kB     00:00
(2/4): rocky-logos-httpd-100.4-7.el10.noarch.rpm                                                                     338 kB/s |  24 kB     00:00
(3/4): nginx-filesystem-1.26.3-1.el10.noarch.rpm                                                                      43 kB/s |  11 kB     00:00
(4/4): nginx-core-1.26.3-1.el10.x86_64.rpm                                                                           989 kB/s | 663 kB     00:00
------------------------------------------------------------------------------------------------------------------------------------------------
Total                                                                                                                545 kB/s | 730 kB     00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                                          1/1
  Running scriptlet: nginx-filesystem-2:1.26.3-1.el10.noarch                                                                                  1/4
  Installing       : nginx-filesystem-2:1.26.3-1.el10.noarch                                                                                  1/4
  Installing       : nginx-core-2:1.26.3-1.el10.x86_64                                                                                        2/4
  Installing       : rocky-logos-httpd-100.4-7.el10.noarch                                                                                    3/4
  Installing       : nginx-2:1.26.3-1.el10.x86_64                                                                                             4/4
  Running scriptlet: nginx-2:1.26.3-1.el10.x86_64                                                                                             4/4

Installed:
  nginx-2:1.26.3-1.el10.x86_64     nginx-core-2:1.26.3-1.el10.x86_64     nginx-filesystem-2:1.26.3-1.el10.noarch     rocky-logos-httpd-100.4-7.el10.noarch

Complete!
root@rockyserver:~# systemctl start nginx
root@rockyserver:~# systemctl enable nginx
Created symlink '/etc/systemd/system/multi-user.target.wants/nginx.service' → '/usr/lib/systemd/system/nginx.service'.
root@rockyserver:~# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
     Active: active (running) since Sun 2026-03-01 01:05:39 IST; 51s ago
 Invocation: 6a493e93cc144f849c10da822f524cf6
   Main PID: 41950 (nginx)
      Tasks: 3 (limit: 31073)
     Memory: 3.1M (peak: 3.2M)
        CPU: 43ms
     CGroup: /system.slice/nginx.service
             ├─41950 "nginx: master process /usr/sbin/nginx"
             ├─41951 "nginx: worker process"
             └─41952 "nginx: worker process"

Mar 01 01:05:39 rockyserver systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Mar 01 01:05:39 rockyserver nginx[41947]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Mar 01 01:05:39 rockyserver nginx[41947]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Mar 01 01:05:39 rockyserver systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
root@rockyserver:~# _
```

Verify nginx configuration files use - #ls -lh /etc/nginx/ or #tree /etc/nginx
Nginx has default website included in its Webroot file
first verify Webroot location in /etc/nginx/nginx.conf file then do - #ls -lh /usr/share/nginx/
or use - #tree./usr/share/nginx (provide your Webroot location)

```
P-Rocky.10 [Running] - Oracle VirtualBox
File  Machine  View  Input  Devices  Help
root@rockyserver:~# tree /etc/nginx/
/etc/nginx/
├── conf.d
├── default.d
├── fastcgi.conf
├── fastcgi.conf.default
├── fastcgi_params
├── fastcgi_params.default
├── koi-utf
├── koi-win
├── mime.types
├── mime.types.default
├── nginx.conf
├── nginx.conf.default
├── scgi_params
├── scgi_params.default
├── uwsgi_params
├── uwsgi_params.default
└── win-utf

3 directories, 15 files
root@rockyserver:~# tree /usr/share/nginx/
/usr/share/nginx/
├── html
│   ├── 50x.html
│   ├── icons
│   │   └── poweredby.png -> ../../../pixmaps/poweredby.png
│   ├── index.html -> ../../testpage/index.html
│   ├── nginx-logo.png
│   ├── poweredby.png -> nginx-logo.png
│   └── system_noindex_logo.png -> ../../pixmaps/system-noindex-logo.png
└── modules

4 directories, 6 files
root@rockyserver:~# _
```

To host new website there are many types, Here we are using port-based hosting
Create a myweb directory in /usr/share/nginx/
Echo into index.html file created in myweb or it will create if not exists
Then create a file “port-sites.conf” in /etc/nginx/conf.d folder
Use vim or nano and paste the below text into the port-sites.conf file

```
P-Rocky.10 [Running] - Oracle VirtualBox
File  Machine  View  Input  Devices  Help
root@rockyserver:~# mkdir /usr/share/nginx/myweb
root@rockyserver:~# echo "<h1> this is new Nginx Web Server </h1>" >> /usr/share/nginx/myweb/index.html
root@rockyserver:~# nano /etc/nginx/conf.d/port-sites.conf
```

```
  GNU nano 8.1                    /etc/nginx/conf.d/port-sites.conf
server {
    listen 8080;
    server_name localhost;

    root /usr/share/nginx/web1;
    index index.html;
}
```

Place the /usr/share/nginx/myweb directory location in /usr/share/nginx/web1
Then esc → :wq
By default nginx listens on 80 port it is already listening by default website ,so we use 8080 port to make nginx listen/host our website on this specific port

Now configure firewall , #firewall-cmd –list-all → lists all firewall rules
Add http and https services using - #firewall-cmd –permanent –add-service=servicename
By default http listens on 80 port and https listens on 443 port so no need to open ports again
Add 8080 port using - #firewall-cmd –permanent –add-port=port.no/tcp ,
This is because we are using port based hosting and the webserver needs 8080 port to listen to the website
Next reload the firewalld daemon using - #firewall-cmd --reload

```
root@rockyserver:~# firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s17 enp0s8
  sources:
  services: cockpit dhcpv6-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
root@rockyserver:~# firewall-cmd --add-service=http --permanent
success
root@rockyserver:~# firewall-cmd --add-service=https --permanent
success
root@rockyserver:~# firewall-cmd --add-port=8080/tcp --permanent
success
root@rockyserver:~# firewall-cmd --reload
success
root@rockyserver:~# _
```

Now restart Nginx #systemctl restart nginx and check if websites are hosted properly or not locally for that use - #curl -I http://ip-address:port
To know the ip address of your vm use - #ip a or ifconfig

```
root@rockyserver:~# systemctl restart nginx
root@rockyserver:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:91:2a:cd brd ff:ff:ff:ff:ff:ff
    altname enx080027912acd
    inet 10.0.3.15/24 brd 10.0.3.255 scope global dynamic noprefixroute enp0s8
       valid_lft 76521sec preferred_lft 76521sec
    inet6 fd17:625c:f037:3:a00:27ff:fe91:2acd/64 scope global dynamic noprefixroute
       valid_lft 86137sec preferred_lft 14137sec
    inet6 fe80::a00:27ff:fe91:2acd/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: enp0s17: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:63:4f:dc brd ff:ff:ff:ff:ff:ff
    altname enx080027634fdc
    inet 192.168.56.108/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s17
       valid_lft 583sec preferred_lft 583sec
    inet6 fe80::22b0:58de:5906:3dc0/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
root@rockyserver:~# curl -I http://192.168.56.108:80
HTTP/1.1 403 Forbidden
Server: nginx/1.26.3
Date: Sun, 01 Mar 2026 14:44:45 GMT
Content-Type: text/html
Content-Length: 153
Connection: keep-alive

root@rockyserver:~# curl -I http://192.168.56.108:8080
HTTP/1.1 404 Not Found
Server: nginx/1.26.3
Date: Sun, 01 Mar 2026 14:44:53 GMT
Content-Type: text/html
Content-Length: 153
Connection: keep-alive

root@rockyserver:~# ss -tulnp | grep 80
tcp   LISTEN 0      511          0.0.0.0:8080      0.0.0.0:*    users:(("nginx",pid=5564,fd=6),("nginx",pid=5563,fd=6),("nginx",pid=5562,fd=6))
tcp   LISTEN 0      511          0.0.0.0:80        0.0.0.0:*    users:(("nginx",pid=5564,fd=7),("nginx",pid=5563,fd=7),("nginx",pid=5562,fd=7))
tcp   LISTEN 0      511             [::]:80           [::]:*    users:(("nginx",pid=5564,fd=8),("nginx",pid=5563,fd=8),("nginx",pid=5562,fd=8))
root@rockyserver:~# getenforce
Permissive
root@rockyserver:~# setenforce 0
root@rockyserver:~# systemctl restart nginx
root@rockyserver:~#
```

Check for on which ports which service is listening #ss -tulnp | grep 80 or #netstat -tulnp | grep 80
If found 403 error then it is a file permission error check for Webroot file permission it should be 755
If found 404 error then it means nginx cant find Webroot, check if used correct path in port-sites.conf file

If there is no problem of both then check for SElinux enforce use - #getenforce
if enforcing then use #setenforce 0, temporarily downs SEinux
Now use curl again

Changing file permissions and ownership
#chown -R root:Developers /usr/share/nginx → changes ownerships to user(root) , group(Developers)
#chmod -R 755 /usr/share/nginx/ → changes file permission (use 775 or 755)
#chmod g+s /usr/share/nginx → set SGID (for constistent group ownership)

Now restart nginx - #systemctl restart nginx

```
root@rockyserver:~# ls -ldh /usr/share/nginx/
drwxr-xr-x. 5 root root 46 Mar  1 19:51 /usr/share/nginx/
root@rockyserver:~# ls -lh /usr/share/nginx/
total 0
drwxr-----. 3 root root 127 Mar  1 01:05 html
drwxr-----. 2 root root   6 Feb  7  2025 modules
drwxr-xr-x. 2 root root  24 Mar  1 19:53 myweb
root@rockyserver:~# chown -R root:Developers /usr/share/nginx
root@rockyserver:~# chmod -R 755 /usr/share/nginx
root@rockyserver:~# chmod g+s /usr/share/nginx/
root@rockyserver:~# ls -ldh /usr/share/nginx/
drwxr-sr-x. 5 root Developers 46 Mar  1 19:51 /usr/share/nginx/
root@rockyserver:~# ls -lh /usr/share/nginx/
total 0
drwxr-xr-x. 3 root Developers 127 Mar  1 01:05 html
drwxr-xr-x. 2 root Developers   6 Feb  7  2025 modules
drwxr-xr-x. 2 root Developers  24 Mar  1 19:53 myweb
root@rockyserver:~# systemctl restart nginx
root@rockyserver:~# curl -I http://192.168.56.108:80
HTTP/1.1 200 OK
Server: nginx/1.26.3
Date: Sun, 01 Mar 2026 15:18:08 GMT
Content-Type: text/html
Content-Length: 7620
Last-Modified: Tue, 26 Aug 2025 00:00:00 GMT
Connection: keep-alive
ETag: "68acf900-1dc4"
Accept-Ranges: bytes
```

```
root@rockyserver:~# chmod -R 775 /usr/share/nginx/
root@rockyserver:~# ls -lh /usr/share/nginx/
total 0
drwxrwxr-x. 3 root Developers 127 Mar  1 01:05 html
drwxrwxr-x. 2 root Developers   6 Feb  7  2025 modules
drwxrwxr-x. 2 root Developers  24 Mar  1 19:53 myweb
root@rockyserver:~#
```

Note: Check the /etc/nginx/conf.d/port-sites.conf

```
  GNU nano 8.1                    /etc/nginx/conf.d/port-sites.conf                    Modified
server {
    listen 8080;
    server_name localhost;

    root /usr/share/nginx/myweb;
    index index.html;
}
```

Now curl the websites again you will get output as “200 ok”

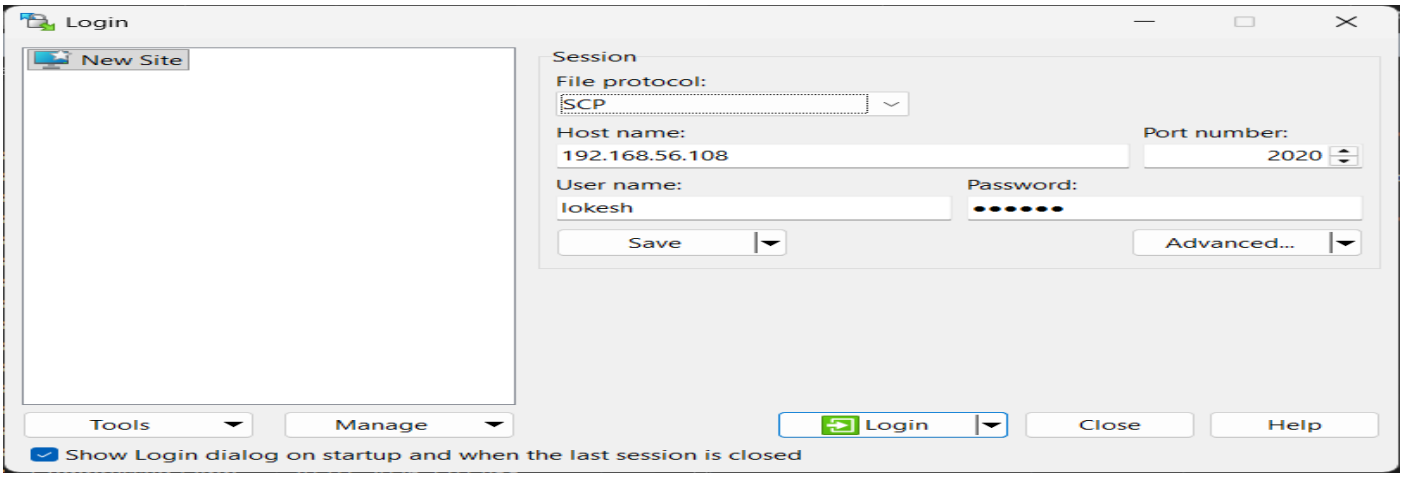
```
root@rockyserver:~# systemctl restart nginx
root@rockyserver:~# curl -I http://192.168.56.108:8080
HTTP/1.1 200 OK
Server: nginx/1.26.3
Date: Sun, 01 Mar 2026 15:25:46 GMT
Content-Type: text/html
Content-Length: 40
Last-Modified: Sun, 01 Mar 2026 14:23:38 GMT
Connection: keep-alive
ETag: "69a44bea-28"
Accept-Ranges: bytes

root@rockyserver:~# curl -I http://192.168.56.108:80
HTTP/1.1 200 OK
Server: nginx/1.26.3
Date: Sun, 01 Mar 2026 15:25:52 GMT
Content-Type: text/html
Content-Length: 7620
Last-Modified: Tue, 26 Aug 2025 00:00:00 GMT
Connection: keep-alive
ETag: "68acf900-1dc4"
Accept-Ranges: bytes

root@rockyserver:~#
```

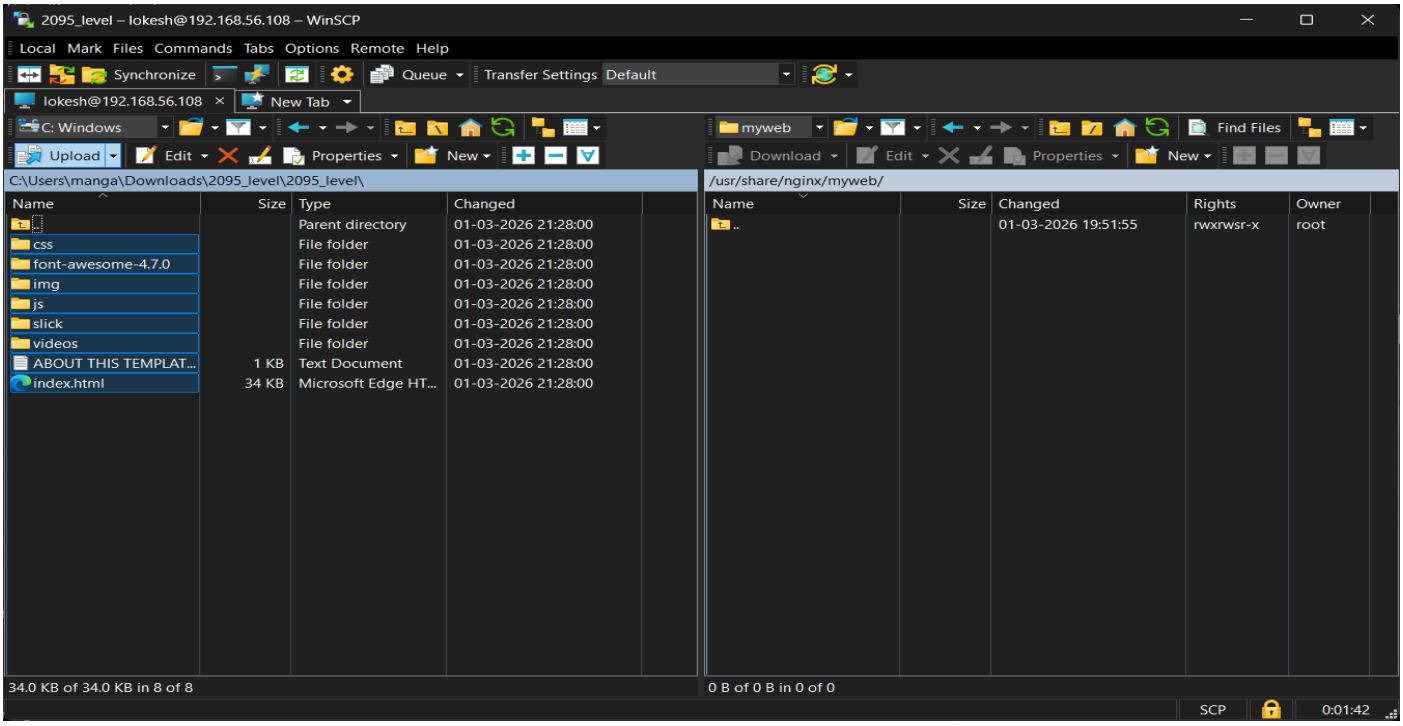
Now to host a website, you have the files in your local machine so, to copy files from local host to remote host (i.e., our server) use winscp tool for that first we need to configure some settings in ssh service
edit /etc/ssh/sshd_config → set 'Port 2020' & 'PermitRootLogin no'
Now allow port 2020 to listen using - #firewall-cmd --add-port=2020/tcp –permanent
restart firewall - #firewall-cmd --reload

```
root@rockyserver:~# ls /etc/ssh
moduli      ssh_config.d  sshd_config.d       ssh_host_ecdsa_key.pub  ssh_host_ed25519_key.pub  ssh_host_rsa_key.pub
ssh_config  sshd_config   ssh_host_ecdsa_key  ssh_host_ed25519_key    ssh_host_rsa_key
root@rockyserver:~# #nano /etc/ssh/sshd_config
root@rockyserver:~# firewall-cmd --add-port=2020/tcp --permanent
success
root@rockyserver:~# firewall-cmd --reload
success
root@rockyserver:~# firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s17 enp0s8
  sources:
  services: cockpit dhcpv6-client http https ssh
  ports: 8080/tcp 2020/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
root@rockyserver:~#
```

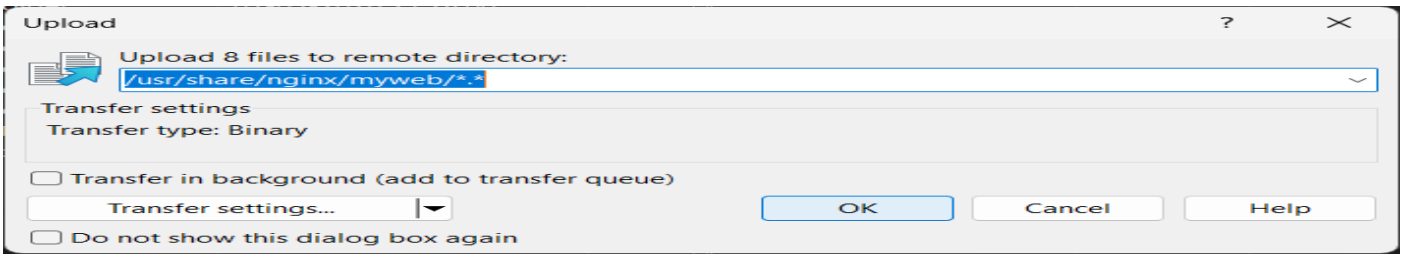
```
  GNU nano 8.1                                                    /etc/ssh/sshd_config
#       $OpenBSD: sshd_config,v 1.104 2021/07/02 05:11:21 dtucker Exp $

# This is the sshd server system-wide configuration file.  See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/usr/local/sbin:/usr/sbin

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented.  Uncommented options override the
# default value.

# To modify the system-wide sshd configuration, create a  *.conf  file under
#  /etc/ssh/sshd_config.d/  which will be automatically included below
Include /etc/ssh/sshd_config.d/*.conf

# If you want to change the port on a SELinux system, you have to tell
# SELinux about this change.
# semanage port -a -t ssh_port_t -p tcp #PORTNUMBER
#
Port 2020
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

^G Help      ^O Write Out   ^F Where Is    ^K Cut       ^T Execute   ^C Location  M-U Undo     M-A Set Mark   M-] To Bracket  M-B Previous
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy       ^B Where Was    M-F Next
```

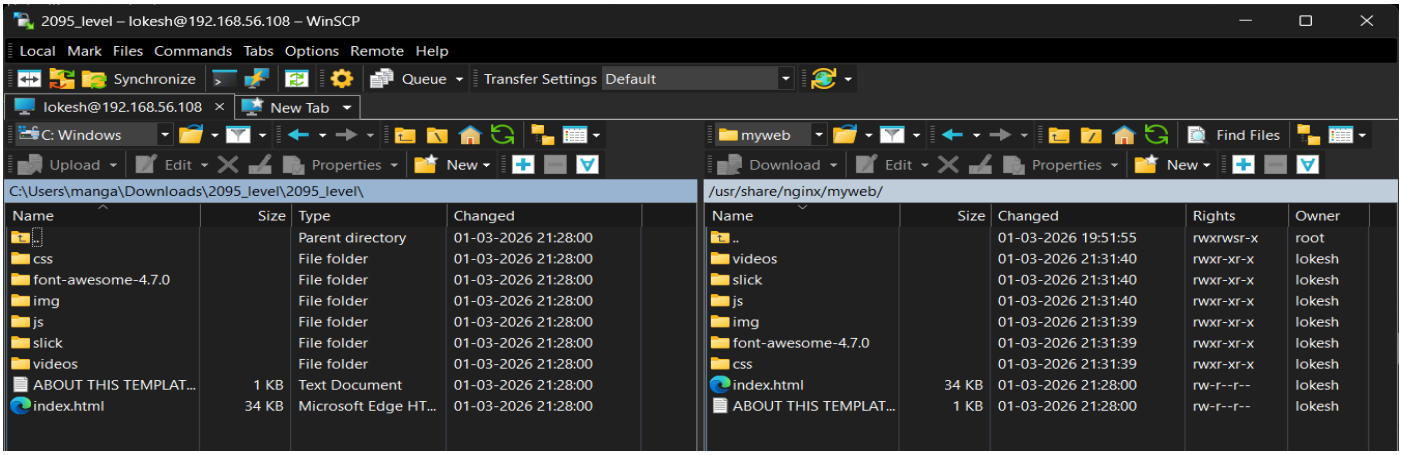
Now use #scp -p 2020 user@ipaddress:/sourcefiles user@ipaddress2:/destinationlocation
Or tools like Winscp, MobaXterm etc.,
Download & open Winscp → select - scp and enter - ipaddress ,port ,username , password
Then click on Login





Select files to upload, Click on upload and then choose directory (eg., nginx webroot), ckick on ok

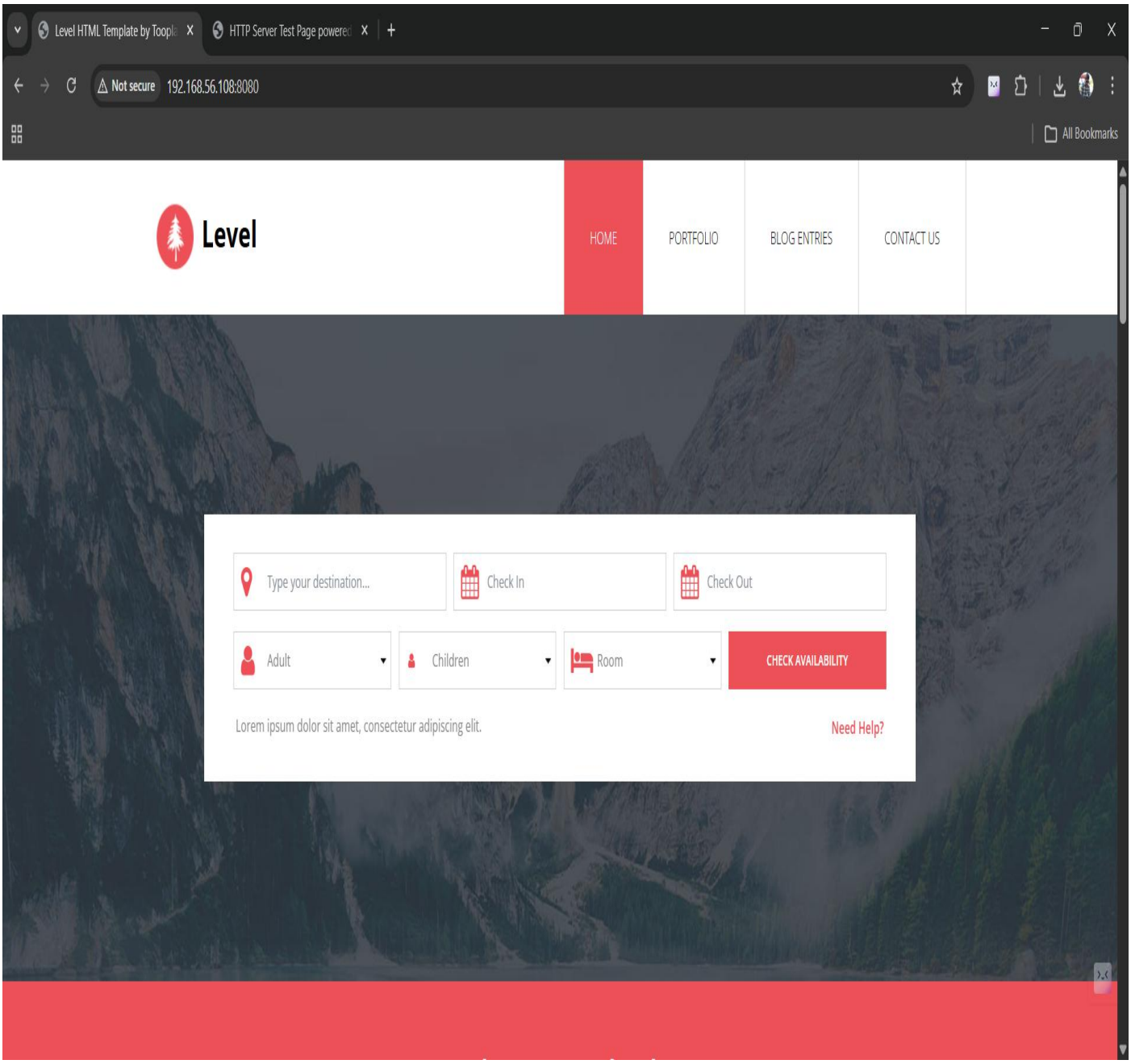




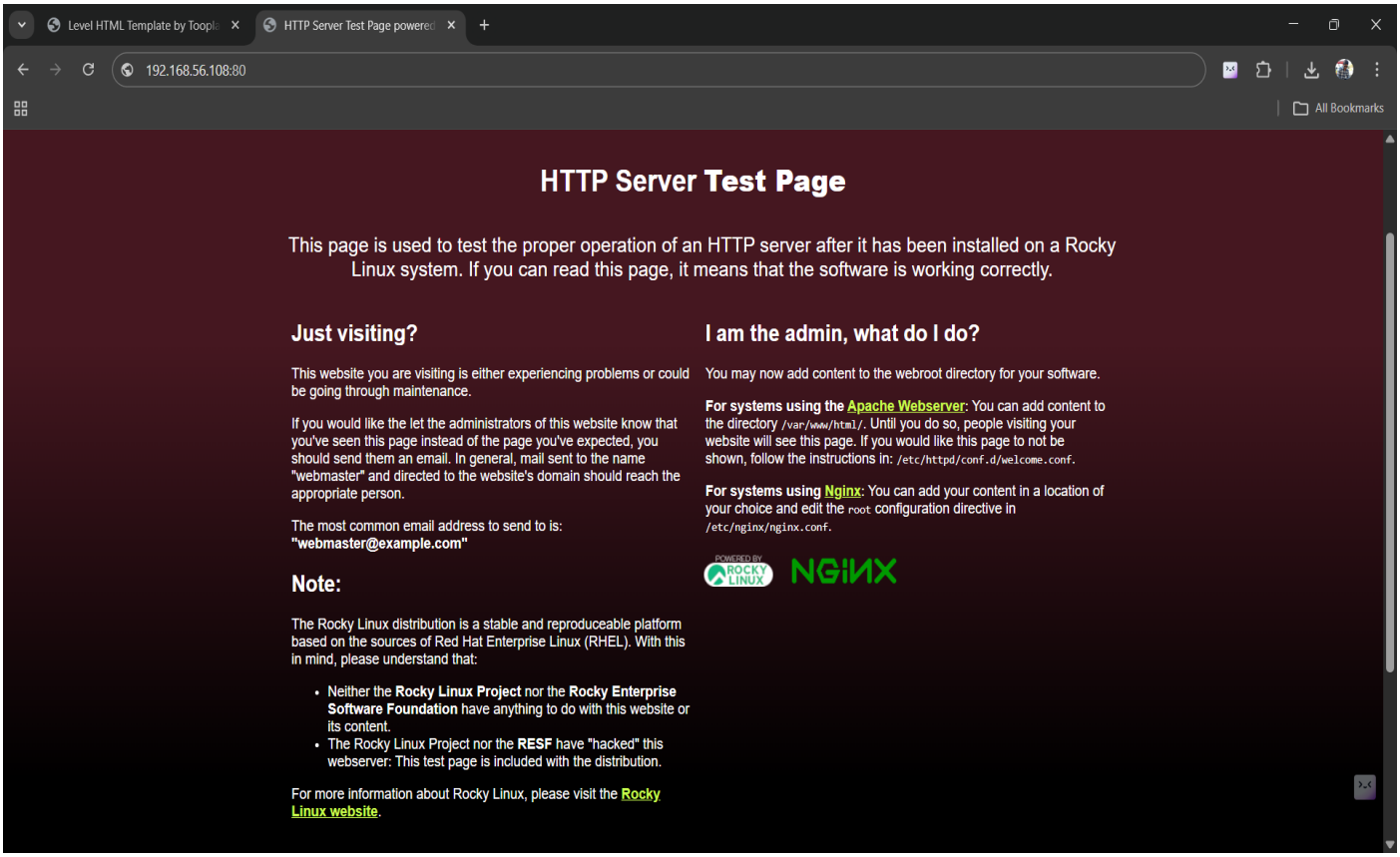
Now restart the nginx server again - #systemctl restart nginx
verify using - #curl -I http:ipaddress:8080

```
root@rockyserver:~# ls /usr/share/nginx/myweb
'ABOUT THIS TEMPLATE.txt'  css  font-awesome-4.7.0  img  index.html  js  slick  videos
root@rockyserver:~# systemctl restart nginx
root@rockyserver:~# curl -I http://192.168.56.108:8080
HTTP/1.1 200 OK
Server: nginx/1.26.3
Date: Sun, 01 Mar 2026 16:03:20 GMT
Content-Type: text/html
Content-Length: 34448
Last-Modified: Sun, 01 Mar 2026 15:58:00 GMT
Connection: keep-alive
ETag: "69a46208-8690"
Accept-Ranges: bytes

root@rockyserver:~#
```

Now Browse the ipaddress:port in any browser
http:192.168.56.108:8080 → this is our custom website hosting



http://192.168.56.108:80 → this is nginx default website



Log Monitoring and Backup

SSH log files are stored in /var/log/secure file.
Nginx log files are stored in /var/log/nginx directory.
Now create a directory and two scripts in it use - #mkdir dirname
Change directory to that folder using – cd /directory
#nano backup.sh → script for backup goes into this file
#nano log_monitor.sh → script for logs goes into this file

```
root@rockyserver:/# cd ..
root@rockyserver:/# pwd
/
root@rockyserver:/# ls /var/log/secure
/var/log/secure
root@rockyserver:/# ls /var/log/nginx
access.log  error.log
root@rockyserver:/# mkdir nginx-logs-backup
root@rockyserver:/# cd nginx-logs-backup
root@rockyserver:/nginx-logs-backup# #nano backup.sh
root@rockyserver:/nginx-logs-backup# #nano log_monitor.sh
root@rockyserver:/nginx-logs-backup# _
```

Change the ownership of directory and its contents using #chown -R username:groupname /dirname
Change file permissions and dir permissions using - #chmod -R 775 /dirname
Verify using - #ls -lh command

```
root@rockyserver:/nginx-logs-backup# ls
backup.sh  log_monitor.sh
root@rockyserver:/nginx-logs-backup# chmod -R 775 /nginx-logs-backup
root@rockyserver:/nginx-logs-backup# ls -lh
total 8.0K
-rwxrwxr-x. 1 lokesh lokesh 463 Mar  1 16:34 backup.sh
-rwxrwxr-x. 1 lokesh lokesh 735 Mar  1 14:57 log_monitor.sh
root@rockyserver:/nginx-logs-backup# chown -R root:Developers /nginx-logs-backup
root@rockyserver:/nginx-logs-backup# ls -lh
total 8.0K
-rwxrwxr-x. 1 root Developers 463 Mar  1 16:34 backup.sh
-rwxrwxr-x. 1 root Developers 735 Mar  1 14:57 log_monitor.sh
root@rockyserver:/nginx-logs-backup#
```

Paste this script into backup.sh file
To execute this script make it executable and then use command - #bash backup.sh or #./backup.sh

```
GNU nano 8.1                                                    backup.sh
#!/bin/bash

WEB_ROOT="/usr/share/nginx/*"

# backup directory
BACKUP_DIR="/nginx-logs-backup/backup"

DATE=$(date +%Y-%m-%d_%H-%M-%S)

#backup file name
BACKUP_FILE="nginx_backup_$DATE.tar.gz"

# Create backup directory if not exists
mkdir -p $BACKUP_DIR

#compress and archive backup
tar -czf $BACKUP_DIR/$BACKUP_FILE $WEB_ROOT
echo "backup completed"

# Delete older backups
find $BACKUP_DIR -name "*.tar.gz" -type f -mtime +7 -delete

echo "Older than 7 days backups are deleted"
```

Paste this script into log_monitor.sh file
To execute this script make it executable and then use command - #bash log_monitor.sh or #./log_monitor.sh

```
GNU nano 8.1                                                    log_monitor.sh
#!/bin/bash
#Source Files
SECURE_LOG="/var/log/secure" #ssh logfile
ACCESS_LOG="/var/log/nginx/access.log" #nginx access.log
ERROR_LOG="/var/log/nginx/error.log" #nginx error.log

DATE=$(date +%Y-%m-%d_%H-%M-%S)

#Output directory
OUTPUT_DIR="/nginx-logs-backup/log_monitor"
#create directory if not exists
mkdir -p $OUTPUT_DIR

OUTPUT_FILE="$OUTPUT_DIR/log-$DATE.txt"

echo "===== Log Report - $DATE =====" >> $OUTPUT_FILE

# Count failed login attempts
FAILED_LOGINS=$(grep "Failed password" $SECURE_LOG | wc -l)
echo "Failed Login Attempts: $FAILED_LOGINS" >> $OUTPUT_FILE

# Count 404 errors
ERROR_404=$(grep " 404 " $ACCESS_LOG | wc -l)
echo "404 Errors: $ERROR_404" >> $OUTPUT_FILE

# Count total errors in error log
ERROR_COUNT=$(wc -l < $ERROR_LOG )
echo "Total Error Log Lines: $ERROR_COUNT" >> $OUTPUT_FILE

echo "Report saved to $OUTPUT_FILE"
```

```
root@rockyserver:/# cd /nginx-logs-backup/
root@rockyserver:/nginx-logs-backup# bash backup.sh
tar: Removing leading `/' from member names
tar: Removing leading `/' from hard link targets
backup completed
Older than 7 days backups are deleted
root@rockyserver:/nginx-logs-backup# ls
backup  backup.sh  log_monitor.sh
root@rockyserver:/nginx-logs-backup# ls -lh backup
total 7.6M
-rw-r--r--. 1 root root 7.6M Mar  1 22:28 nginx_backup_2026-03-01_22-28-13.tar.gz
root@rockyserver:/nginx-logs-backup# bash log_monitor.sh
Report saved to /nginx-logs-backup/log_monitor/log-2026-03-01_22-29-24.txt
root@rockyserver:/nginx-logs-backup# ls
backup  backup.sh  log_monitor  log_monitor.sh
root@rockyserver:/nginx-logs-backup# ls -lh log_monitor
total 4.0K
-rw-r--r--. 1 root root 111 Mar  1 22:29 log-2026-03-01_22-29-24.txt
```

Now to automate this scripts execution use - #crontab -e
Paste the scripts location here and configure time
Example: run backup.sh script every day at 2'0 clock
run log_monitor.sh script every 15 minutes

```
*/15 * * * * /nginx-logs-backup/log_monitor.sh
* 2 * * * /nginx-logs-backup/backup.sh
~
~
~
~
```

Monitoring

Monitor system resources using - #top and #htop, press "q" to exit from top
Monitor running processes using - #ps -ef , #ps -aux
Check the diskspace using - #df -Th
Check the directory storage using - #du /dir_name

```
top - 22:23:40 up  4:54,  2 users,  load average: 0.06, 0.02, 0.00
Tasks: 153 total,   1 running, 152 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.2 sy,  0.0 ni, 98.7 id,  0.0 wa,  1.0 hi,  0.2 si,  0.0 st
MiB Mem :   4655.4 total,   4008.4 free,    521.4 used,    370.5 buff/cache
MiB Swap:   5040.0 total,   5040.0 free,      0.0 used.   4134.0 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   6713 root      20   0   10220   5488   3296 R   0.3   0.1   0:00.12 top
      1 root      20   0   50020  41844  10524 S   0.0   0.9   0:19.23 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.07 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
     11 root      20   0       0      0      0 I   0.0   0.0   0:00.00 kworker/u8:0-events_unbound
     12 root      20   0       0      0      0 I   0.0   0.0   0:00.22 kworker/u8:1-ipv6_addrconf
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     16 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
     17 root      20   0       0      0      0 S   0.0   0.0   0:00.63 ksoftirqd/0
     18 root      20   0       0      0      0 I   0.0   0.0   0:05.15 rcu_preempt
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
     20 root      20   0       0      0      0 S   0.0   0.0   0:00.11 rcu_exp_gp_kthread_worker
     21 root      rt   0       0      0      0 S   0.0   0.0   0:00.25 migration/0
     22 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
     23 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     24 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
     25 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/1
     26 root      rt   0       0      0      0 S   0.0   0.0   0:00.76 migration/1
     27 root      20   0       0      0      0 S   0.0   0.0   0:00.61 ksoftirqd/1
     29 root       0 -20       0      0      0 I   0.0   0.0   0:00.66 kworker/1:0H-kblockd
     34 root      20   0       0      0      0 S   0.0   0.0   0:00.01 kdevtmpfs
     35 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-inet_frag_wq
     37 root      20   0       0      0      0 S   0.0   0.0   0:00.10 kauditd
     38 root      20   0       0      0      0 S   0.0   0.0   0:00.05 khungtaskd
     39 root      20   0       0      0      0 S   0.0   0.0   0:00.00 oom_reaper
     41 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-writeback
     42 root      20   0       0      0      0 S   0.0   0.0   0:01.05 kcompactd0
     43 root      25   5       0      0      0 S   0.0   0.0   0:00.00 ksmd
     44 root      39  19       0      0      0 S   0.0   0.0   0:06.77 khugepaged
     45 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-cryptd
     46 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kblockd
     47 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-blkcg_punt_bio
     48 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kintegrityd
     49 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 irq/9-acpi
     51 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-tpm_dev_wq
     52 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-md
```

To kill a process use - #kill pid or #kill processname or #killall, To free up resources

```
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 17:29 ?        00:00:19 /usr/lib/systemd/systemd --switched-root --system --deserialize=43 rhgb
root           2       0  0 17:29 ?        00:00:00 [kthreadd]
root           3       2  0 17:29 ?        00:00:00 [pool_workqueue_release]
root           4       2  0 17:29 ?        00:00:00 [kworker/R-rcu_gp]
root           5       2  0 17:29 ?        00:00:00 [kworker/R-sync_wq]
root           6       2  0 17:29 ?        00:00:00 [kworker/R-slub_flushwq]
root           7       2  0 17:29 ?        00:00:00 [kworker/R-netns]
root          10       2  0 17:29 ?        00:00:00 [kworker/0:0H-events_highpri]
root          11       2  0 17:29 ?        00:00:00 [kworker/u8:0-events_unbound]
root          12       2  0 17:29 ?        00:00:00 [kworker/u8:1-ipv6_addrconf]
root          13       2  0 17:29 ?        00:00:00 [kworker/R-mm_percpu_wq]
root          14       2  0 17:29 ?        00:00:00 [rcu_tasks_kthread]
root          15       2  0 17:29 ?        00:00:00 [rcu_tasks_rude_kthread]
root          16       2  0 17:29 ?        00:00:00 [rcu_tasks_trace_kthread]
root          17       2  0 17:29 ?        00:00:00 [ksoftirqd/0]
root          18       2  0 17:29 ?        00:00:05 [rcu_preempt]
root          19       2  0 17:29 ?        00:00:00 [rcu_exp_par_gp_kthread_worker/0]
root          20       2  0 17:29 ?        00:00:00 [rcu_exp_gp_kthread_worker]
root          21       2  0 17:29 ?        00:00:00 [migration/0]
root          22       2  0 17:29 ?        00:00:00 [idle_inject/0]
root          23       2  0 17:29 ?        00:00:00 [cpuhp/0]
root          24       2  0 17:29 ?        00:00:00 [cpuhp/1]
root          25       2  0 17:29 ?        00:00:00 [idle_inject/1]
root          26       2  0 17:29 ?        00:00:00 [migration/1]
root          27       2  0 17:29 ?        00:00:00 [ksoftirqd/1]
root          29       2  0 17:29 ?        00:00:00 [kworker/1:0H-kblockd]
root          34       2  0 17:29 ?        00:00:00 [kdevtmpfs]
root          35       2  0 17:29 ?        00:00:00 [kworker/R-inet_frag_wq]
root          37       2  0 17:29 ?        00:00:00 [kauditd]
root          38       2  0 17:29 ?        00:00:00 [khungtaskd]
root          39       2  0 17:29 ?        00:00:00 [oom_reaper]
root          41       2  0 17:29 ?        00:00:00 [kworker/R-writeback]
root          42       2  0 17:29 ?        00:00:01 [kcompactd0]
root          43       2  0 17:29 ?        00:00:00 [ksmd]
root          44       2  0 17:29 ?        00:00:06 [khugepaged]
root          45       2  0 17:29 ?        00:00:00 [kworker/R-cryptd]
root          46       2  0 17:29 ?        00:00:00 [kworker/R-kblockd]
root          47       2  0 17:29 ?        00:00:00 [kworker/R-blkcg_punt_bio]
root          48       2  0 17:29 ?        00:00:00 [kworker/R-kintegrityd]
root          49       2  0 17:29 ?        00:00:00 [irq/9-acpi]
root          51       2  0 17:29 ?        00:00:00 [kworker/R-tpm_dev_wq]
root          52       2  0 17:29 ?        00:00:00 [kworker/R-md]
root          53       2  0 17:29 ?        00:00:00 [kworker/R-md_bitmap]
root          54       2  0 17:29 ?        00:00:00 [kworker/R-edac-poller]
root          55       2  0 17:29 ?        00:00:00 [watchdogd]
root          58       2  0 17:29 ?        00:00:00 [kswapd0]
root          66       2  0 17:29 ?        00:00:00 [kworker/R-kthrotld]
root          71       2  0 17:29 ?        00:00:00 [kworker/R-acpi_thermal_pm]
:
```

```
root@rockyserver:/# df -Th
Filesystem                  Type      Size  Used Avail Use% Mounted on
/dev/mapper/rl_rockyserver-root xfs    64G  3.2G   61G   5% /
devtmpfs                    devtmpfs  2.3G     0  2.3G   0% /dev
tmpfs                       tmpfs     2.3G     0  2.3G   0% /dev/shm
tmpfs                       tmpfs     932M   33M  899M   4% /run
/dev/sda2                   xfs       960M  426M  535M  45% /boot
/dev/mapper/rl_rockyserver-home xfs    31G  637M   31G   3% /home
tmpfs                       tmpfs     1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs                       tmpfs     466M  4.0K  466M   1% /run/user/0
tmpfs                       tmpfs     1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs                       tmpfs     1.0M     0  1.0M   0% /run/credentials/getty@tty2.service
```

```
root@rockyserver:/nginx-logs-backup# du /nginx-logs-backup/
7728    /nginx-logs-backup/backup
8       /nginx-logs-backup/log_monitor
7744    /nginx-logs-backup/
root@rockyserver:/nginx-logs-backup# du -h /nginx-logs-backup/
7.6M    /nginx-logs-backup/backup
8.0K    /nginx-logs-backup/log_monitor
7.6M    /nginx-logs-backup/
root@rockyserver:/nginx-logs-backup#
```